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PALEOMAGNETISM OF EARLY CRETACEOUS ARAPEY FORMATION (NORTHERN URUGUAY)

MIGUEL CERVANTES SOLANO[1], AVTO GOGUITCHAICHVILI[1*], LEDA SÁNCHEZ BETTUCCI[2], RUBEN CEJUDO RUIZ[1], MANUEL CALVO-RATHERT[3], VICENTE CARLOS RUIZ-MARTINEZ[4], RUTH SOTO[5] AND LUIS M. ALVA-VALDIVIA[6]

1 Laboratorio Interinstitucional de Magnetismo Natural, Instituto de Geofísica - Sede Michoacán, Universidad Nacional Autónoma de México, Campus Morelia, 58089 Morelia, Mexico (avto@geofisica.unam.mx)

2 Departamento de Geología, Area Geofísica-Geotectónica, Facultad de Ciencias, Universidad de la República, 11200 Montevideo, Uruguay

3 Laboratorio de Paleomagnetismo, Departamento de Física, Escuela Politécnica Superior, Universidad de Burgos, C/Francisco de Vitoria, s/n, 09006, Burgos, Spain

4 Departamento de Física de la Tierra, Astronomía y Astrofísica I, Universidad Complutense de Madrid, 28040 Madrid, Spain

5 Instituto Geológico y Minero de España (IGME), Zaragoza, Spain

6 Laboratorio de Paleomagnetismo, Instituto de Geofísica, Universidad Nacional Autónoma de México, 04510 México DF, Mexico

* Corresponding author

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ABSTRACT

We report detailed rock-magnetic and paleomagnetic directional data from 35 lava flows (302 standard paleomagnetic cores) sampled in the Central-Northern region of Uruguay in order to contribute to the study of the paleosecular variation of the Earth's magnetic field during early Cretaceous and to obtain precise Cretaceous paleomagnetic pole positions for stable South America. The average unit direction is rather precisely determined from 29 out of 35 sites. All A_{95} confidence angles are less than 8°, which points to small within-site dispersion and high directional stability. Normal polarity magnetizations are revealed for 19 sites and 10 are reversely magnetized. Two other sites yield well defined intermediate polarities. The mean direction, supported by a positive reversal test is in reasonably good agreement with the expected paleodirection for Early Cretaceous stable South America and in disagreement with a 10° clockwise rotation found in the previous studies. On the other hand, paleomagnetic poles are significantly different from the pole position suggested by hotspot reconstructions, which may be due to true polar wander or the hotspot motion. Our data suggest a different style of secular variation during (and just before) the Cretaceous Normal Superchron and the last 5 Ma, supporting a link between paleosecular variation and reversal frequency.

Keywords: paleosecular variation, Cretaceous paleomagnetic poles, Paraná Magmatic Province, Uruguay

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1. INTRODUCTION

The variations of the geomagnetic field cover timescales from milliseconds to a few million years and originate in both external and internal regions of the Earth. While relatively short variations (from milliseconds to a few decades) are related to the solar magnetic field, the long-term fluctuations (few years to millions of years) result from the effect of magnetic induction in the fluid outer core and magnetic diffusion in the core and the mantle (*Gubbins and Herrero-Bervera, 2007*). Thus, revealing the variability of the geomagnetic field with time is essential for understanding the conditions in the Earth's liquid core and at the core-mantle boundary.

A generally accepted approach to estimate the paleosecular variation (PSV) consists in observing the angular standard deviation (ASD) of virtual geomagnetic poles (VGPs) for a given locality. Several combinations of dipole and non-dipole components predict the ASD characteristic of PSV with latitude (*McFadden et al., 1988, 1991*; *Lawrence et al., 2006*). It is interesting to mention that *Johnson et al. (2008)* reported a detailed synthesis of a new generation of paleomagnetic data compilations for the last 5 Ma showing that the latitudinal dependence of VGP (virtual geomagnetic poles) scatter for these data appears much less important.

While abundant and well documented paleomagnetic records are available for recent periods (for instance last 10 Ma or so), older periods are still poorly studied in terms of paleosecular variations of the Earth's magnetic field. Recently, *Biggin et al. (2008)* carefully investigated two key intervals: the Cretaceous Normal Superchron (CNS: 84–125 Ma) when the field was dominantly of a single polarity for 40 Ma and the Jurassic (145–200 Ma), when reversals occurred at an average rate of as much as 4.6 Myr^{-1}. These are actually periods of time in which the geomagnetic reversal frequency was dramatically different and thus different styles of secular variation are expected. The transition mode between these two extremely distinct periods, however, is not known. New paleomagnetic data reported here from 35 independent lava flows erupted between 135 and 130 Ma in northern Uruguay may help to elucidate some of these questions.

On the other hand, the Early Cretaceous paleomagnetic pole position for stable South America is still matter of debates. The Paraná Magmatic Province (PMP, Fig. 1), located in southern Brazil, Uruguay, Paraguay and Argentina represents one of the world's largest volumes of Mesozoic continental flood basalts and has an Early Cretaceous age. The PMP has been extensively studied, with more than 300 sites recording the direction of the paleomagnetic field (*Ernesto et al., 1990*). *Alva-Valdivia et al. (2003)* found that all PMP paleomagnetic poles lie about 10° away from the pole predicted by an assumed fixed hotspot reconstruction of South America in contradiction with early studies. We note that some of the paleomagnetic data reported in the latter study (see Table 2 in *Alva-Valdivia et al., 2003*) have a slightly non-Fisherian distribution of VGPs, probably suggesting either non-removal of secondary overprints or differential tectonic disturbance of their sampling sites. Moreover, many studies are based on only three oriented hand samples per site and thus does not suit the quality standard required in modern paleomagnetic studies. The same is true for the study conducted by *Mena et al. (2006)* who only used few (3 to 5) samples on the Argentinean part of PMP (Posadas Formation).

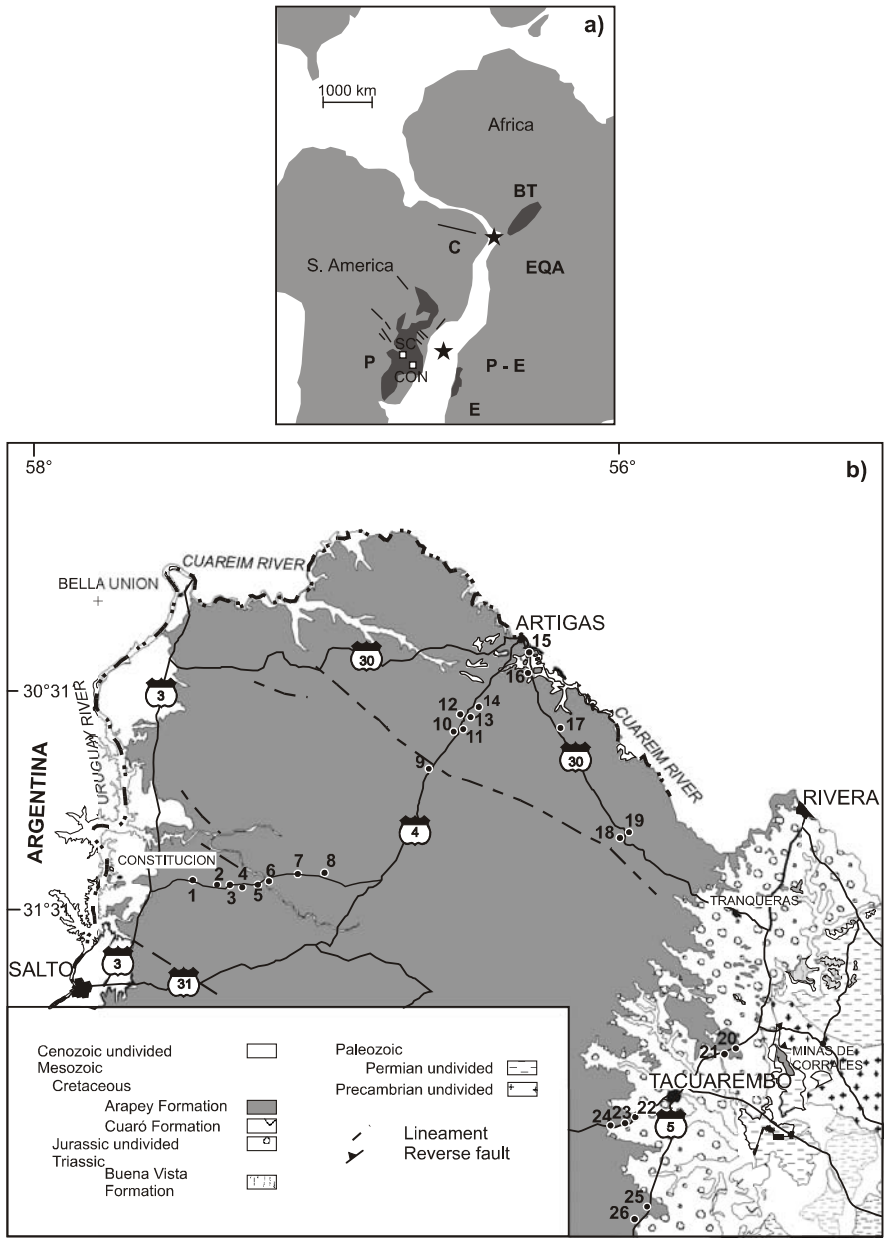


Fig. 1. **a)** Shematic map showing the geographic distribution of Paraná-Etendeka Igneous Province (re-drawn after *Alva-Valdivia et al., 2003*, © Elsevier B.V.); **b)** Simplified geologic map of northern Uruguay, numbers show locations of studied sites.

In this study, we present new detailed rock-magnetic and paleomagnetic directional data from 35 lavas (302 standard paleomagnetic cores) sampled in the Central-Northern region of Uruguay in order to contribute to both paleosecular variation of Earth's magnetic field during early Cretaceous and paleomagnetic pole positions for the stable south America. This work comes to complete the reconnaissance paper (first sample collection campaign in Uruguay in 2005) of *Goguitchaichvili et al. (2008)*. These authors reported five absolute paleointensity results and no directional data. The additional samples obtained in 2007 were decisive to precisely estimate both paleosecular variation and robust paleomagnetic pole.

2. GEOLOGICAL FRAMEWORK AND SAMPLING DETAILS

The Paraná Magmatic Province (PMP) was formed during the continental rifting of Pangea in the Early Cretaceous (Tristan da Cunha mantelic plume after *O'Connor and Duncan, 1990*; *Peate et al., 1990*; *Hawkesworth et al., 1992*) and is part of Paraná-Etendeka Igneous Province (PEIP). The PEIP (Fig. 1a) is one of the main flood volcanic provinces in the world covering an area of ca. 1.2×10^6 km^2, with its magmatic activity peak in ca. 132 Ma (*Erlank et al., 1984*; *Bellieni el al., 1984*; *Renne et al. 1992, 1996a,b*). It was estimated (*Bellieni et al., 1984, 1986*) that the acidic volcanic rocks represent only 3% of the total volume for the South American portion whereas half of African portion (Etendeka) is covered by such type of rocks. This compositional asymmetry seems to be related to the rift geometry (*Turner et al., 1994*).

The Uruguayan part of the PMP, named Arapey Formation by *Bossi (1966)*, outcrops mainly in the NW part of that country (Fig. 1b). Some old K-Ar radiometric ages obtained for this formation range from 127 ± 3 to 153 ± 8 Ma. The most recent ages obtained by *Feraud et al. (1999)* using $^{40}Ar/^{39}Ar$ methodology yielded 132.9 ± 1.3; 132.2 ± 0.5; 131.8 ± 0.4 and 129.9 ± 1.1 Ma. Regarding chemical composition, the Paraná Province basalts display characteristics of bimodality with a strong geographic correlation. The Arapey Formation is part of the so-called low TiO_2 subprovince (sensu *Bellieni et al., 1986*; *Peate, 1997*). The basalts are dark-grey or dark-green olivine basalts slightly oversaturated, with normative hypersthene. The volcanic rocks of Arapey Formation are emplaced above aeolian Jurassic sandstones (Tacuarembó Formation). The lava flows systematically show a tilting of 3° to 5° to the WSW.

In total, we obtained 302 standard paleomagnetic cores belonging to 35 sites (Fig. 1b) distributed along road outcrops of Northern Uruguay during 2005 and 2007 sample collection campaigns. The samples were distributed throughout each flow both horizontally and vertically. All lava flows sampled were almost horizontal (dip less than 4°). In general, samples were obtained with the hope of collecting samples with the finest grain size of material. Cores were obtained with a gasoline-powered portable drill, and then oriented with a magnetic and in most cases also with a sun compass.

3. LABORATORY TECHNIQUES

The remanent magnetization of 7 to 9 samples per site was measured with JR6 spinner magnetometer (AGICO Ltd.) at Laboratorio Interinstitucional de Magnetismo Natural,

(UNAM, Campus Morelia, Mexico) and with a 2G cryogenic magnetometer at the paleomagnetic laboratory of the University of Burgos (Spain). Measurements were recorded after stabilization of the remanence. Both alternating field (AF) demagnetization (mainly) using a Molspin AF-demagnetizer and stepwise thermal demagnetization using a non-inductive Schonstedt furnace were carried out. Low-field susceptibility measurements (κ-T curves) under air were carried out using a Highmoore susceptibility bridge equipped with furnace while hysteresis and isothermal remanence (IRM) acquisition measurements at room temperature were performed on representative samples from all the studied sites (one specimen per site) using the AGFM 'Micromag' apparatus. The saturation remanent magnetization (J_{rs}), the saturation magnetization (J_s) and coercitive force (H_c) were calculated after correction for the paramagnetic contribution. The coercitivity of remanence (H_{cr}) was determined by applying progressively increasing backfield after saturation.

4. PRINCIPAL RESULTS OF MAGNETIC MEASUREMENTS

Prior to magnetic treatments, the viscosity index was determined following procedures described by *Prévot et al. (1983)*. This allows estimation of the capacity of a sample to acquire a viscous remanent magnetization, and is therefore useful to obtain information about its paleomagnetic stability. Three samples from each site were subjected to these experiments and although viscosity indexes varied between 0 and 42.7, most values were lower than 10%.

For more than half of the sites, a stable paleomagnetic component was retrieved (Fig. 2, sample 95A007A). A minor secondary component was easily removed applying 10 mT peak alternative field. The median destructive fields (*MDF*) for these samples ranges mostly from 35 to 45 mT, suggesting 'small' pseudo-single domain grains as remanent magnetization carriers (*Dunlop and Özdemir, 1997*). Other samples are characterized by strong secondary overprints (sample 95U011B, Fig. 2) which are removed at 20 mT. Few sites are characterized by unusually high intensity (more than 30 A/m, Fig. 2, sample 95A023) and scattered natural remanent magnetization (*NRM*) directions. Both factors point to a possible strong lightning-produced magnetization overprint. However, AF demagnetization easily revealed the primary, characteristic paleodirection.

In this section we will only recall some main results of our rock-magnetic experiments since these samples were already partly investigated by *Goguitchaichvili et al. (2008)*. The susceptibility vs. temperature experiments indicates, in most cases, the presence of Ti-poor titanomagnetites. So do microscopic observations on polished sections, which show that the main magnetic mineral is low-Ti titanomagnetite associated with ilmenite and hematite exsolution of trellis or composite texture (after *Haggerty, 1976*), probably formed as a result of deuteric oxidation of titanomagnetite during the initial flow cooling. Judging from the ratios of hysteresis parameters (H_{cr}/H_c range, 1.72 and 2.97 and J_{rs}/J_s range, 0.14 to 0.29), it seems that all samples fall in the pseudo-single domain (PSD) grain size region (*Day et al., 1977*), probably indicating a mixture of multidomain (MD) and a significant amount of single domain (SD) grains. Isothermal remanent magnetization (IRM) acquisition curves (Fig. 3) were found very similar for all samples. Saturation is

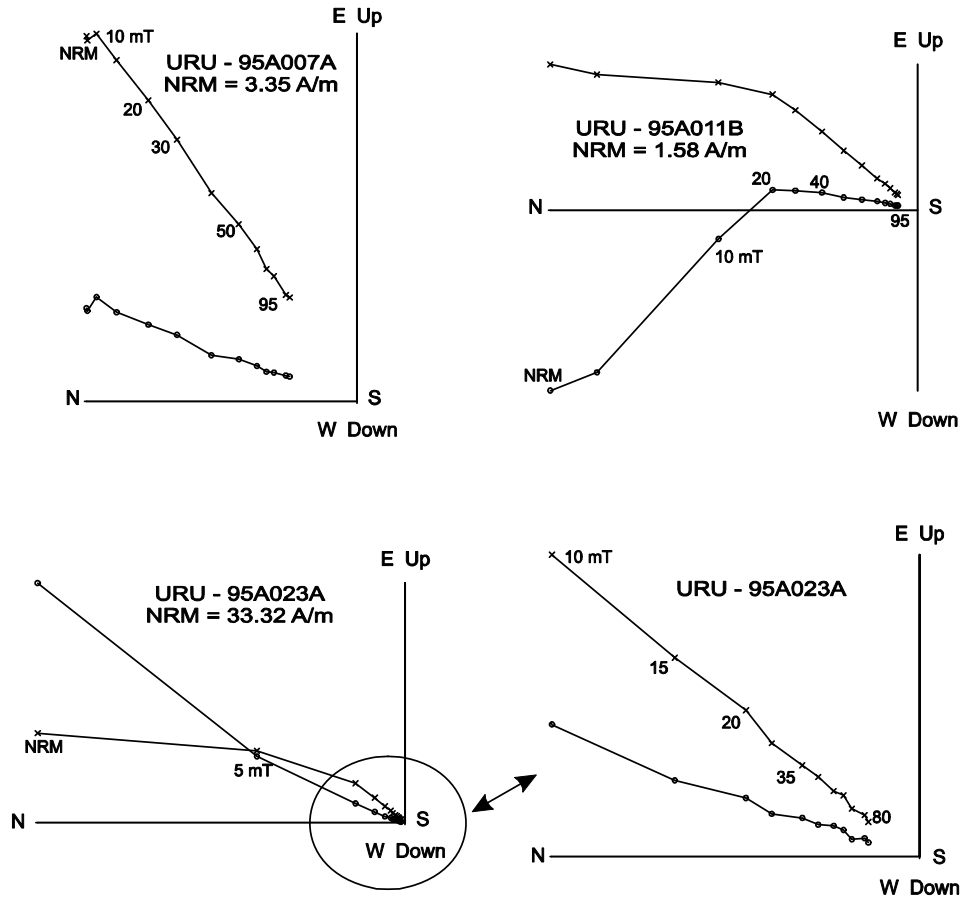


Fig. 2. Orthogonal vector plots of stepwise alternating field demagnetization of representative samples (stratigraphic coordinates). The numbers refer to the peak alternating fields in mT. ○ - projections into the horizontal plane, × - projections into the vertical plane.

reached in moderate fields of the order of 100–200 mT, which points to some spinel as remanence carrier. The sample 05A178 (single case) remains unsaturated until 300 mT (Fig. 3). This may due either the presence of dominantly single-domain titanomagnetite grains or some mixture of titanomagnetites and titanohematites.

5. MAIN RESULTS AND DISCUSSION

The characteristic magnetization direction was determined by the least squares method (*Kirschvink, 1980*), 5 to 11 points being taken in the principal component analysis for this determination. Directions were averaged by unit and the statistical parameters were calculated assuming a Fisherian distribution. The average unit directions are rather

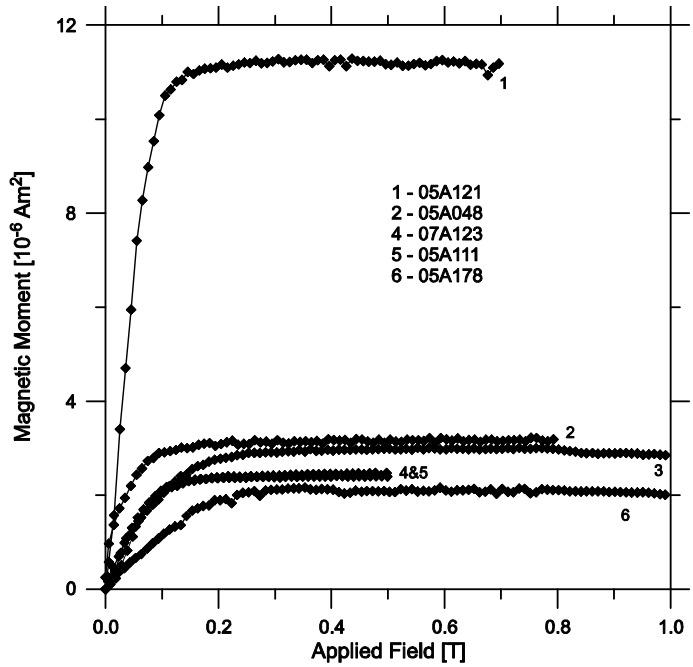


Fig. 3. Typical examples of isothermal remanence acquisition curves of small chip samples from the studied volcanic units.

precisely determined (Table 1, Fig. 4) for 29 out of 35 sites. All α_{95} are less than 8° which points to small within-site dispersion and high directional stability. Two sites (AR09 and UR27) yielded unusually high dispersion yielding α_{95} equal to 13.1° and 18.1°, respectively. No paleodirections were determined for sites AR14 and UR31 because of their very complex behavior during the paleomagnetic treatments. An intermediate polarity magnetization (using the VGP latitude of 45° as cut-off angle after *Johnson et al., 2008*) seems to be present in sites UR25 (paleolatitude is −15.5°) and UR32 (−41.3°). It should be pointed out that these directions are quite precisely determined (Table 1) and most probably have a geomagnetic significance. These sites, underlined in Fig. 4 and Table 1 were rejected for the following paleomagnetic analysis.

Nineteen sites give normal polarity magnetization and 10 are reversely magnetized. The mean paleomagnetic direction of normal polarity sites is $I = -41.9°$, $D = 357.8°$, $k = 70$, $\alpha_{95} = 4.1°$, $N = 19$ while reversely magnetized sites give $I = 48.5°$, $D = 177.3°$, $k = 29$, $\alpha_{95} = 9.2°$, $N = 10$. These results point to *almost* antipodal mean directions, since the reversal test as defined by *McFadden and McElhinny (1990)* is positive corresponding to type B. Thus, the hypothesis of a common mean direction may not be rejected at the 95% level. The mean paleomagnetic direction obtained in this study using all the 29 accepted sites is $I = 44.2°$, $D = 177.6°$, $k = 46$, $\alpha_{95} = 4.0°$ and the corresponding mean paleomagnetic pole position is $P_{lat} = -84.8°$, $P_{long} = 95.8°$, $K = 42$, $\alpha_{95} = 4.2°$.

Table 1. Paleomagnetic results of two studied sections from Paraná. Lat. and Long. are the geographic co-ordinates of each lava flow; *n/N* is the number of used/treated samples; Inc. and Dec. is the inclination and declination of the site-mean *ChRM*; *k* and α_{95} are the precision parameter and the confidence cone of the Fisher statistics. P_{lat}, P_{long} is the latitude and longitude of the virtual geomagnetic pole, respectively. * refers to the sites yielding α_{95} above 10° while ** indicates the sites with apparently intermediate paleodirections.

Site	Lat. [°]	Long. [°]	*n*	*N*	Inc. [°]	Dec [°]	*k*	α_{95} [°]	P_{lat} [°]	P_{long} [°]
AR01 (1)	–31.32	–57.14	8	8	–51.7	10.9	1578	1.4	–80.7	–138.0
AR02 (2)	–31.33	–57.33	7	8	–31.1	349.9	93	6.3	–72.8	88.1
AR03 (3)	–31.31	–57.25	7	8	–36.8	10.2	363	3.2	–75.8	165.4
AR04 (4)	–31.3	–57.23	8	8	–47.8	358.2	1348	1.5	–87.1	89.6
AR05 (5)	–31.28	–57.13	8	8	–48.7	354.3	91	5.8	–84.8	49.8
AR06 (6)	–31.24	–57.1	7	8	–59.9	335.6	98	6.2	–68.2	0.2
AR07 (7)	–31.12	–56.99	8	8	–39.7	352.3	111	5.3	–79.0	82.5
AR08 (8)	–31.2	–56.89	8	8	–40.8	359.6	142	4.6	–82.1	120.4
AR09* (9)	–30.76	–56.76	3	8	–54.2	336.2	83	13.1	–69.6	15.7
AR10 (10)	–30.65	–56.67	8	8	46.8	179.6	381	2.8	–87.4	115.6
AR11 (11)	–30.66	–56.67	8	8	52.9	169.3	244	3.6	–80.5	13.3
AR12 (12)	–30.62	–56.66	7	7	46.2	172.6	78	6.9	–82.8	57.0
AR13 (13)	–30.62	–56.66	6	8	48.1	164.6	196	5.4	–76.6	35.8
AR14*(14)	–30.59	–56.62	1	8	49.1	163.5	–	–	–	–
AR15 (15)	–30.43	–56.45	8	8	42.2	168.6	163	4.3	–78.2	61.7
AR16 (16)	–30.53	–56.43	8	8	38.6	171.2	259	3.4	–78.2	79.5
AR17 (17)	–30.7	–56.33	8	8	–42.4	13.6	51.7	7.9	–76.5	189.9
AR18 (18)	–31.14	–55.92	7	9	56.1	173.5	112	5.6	–82.3	–13.3
AR19 (19)	–31.14	–55.91	7	8	28.1	219.6	269	3.7	–50.4	199.2
AR20 (20)	–31.59	–55.77	8	8	–48.5	356.9	1109	1.8	–86.6	71.8
AR21 (21)	–31.6	–55.81	8	8	–32.9	353.2	128	4.9	–75.0	98.4
AR22 (22)	–31.81	–56.22	8	8	–37.4	357.6	194	3.9	–78.9	112.1
AR23 (23)	–31.81	–56.22	7	8	–36.2	353.6	56	8.1	–77.0	96.1
AR24 (24)	–31.82	–56.3	8	8	–41.8	359.1	118	5.1	–82.2	117.6
AR25 (25)	–32.06	–56.07	8	8	–48.1	2.3	418	2.7	–86.5	158.5
AR26 (26)	–32.18	–56.14	6	8	–29.4	351.3	1469	1.8	–71.8	96.2
UR25**(27)	–31.09	–57.66	7	8	–69.9	107.2	1582	1.5	–15.5	–93.5
UR26 (28)	–30.97	–57.7	8	8	–42.2	356.6	172	4.2	–82.8	96.9
UR27*(29)	–30.52	–57.43	4	8	–41.4	324.5	27	18.1	–57.8	35.9
UR28 (30)	–30.62	–56.82	6	8	–39.1	355.3	423	3.8	–80.5	95.8
UR29 (31)	–30.51	–56.69	5	8	62.5	164.7	102	7.6	–72.0	–18.7
UR30 (32)	–30.57	–56.33	4	8	51.5	176.8	224	6.2	–86.8	2.7
UR31*(33)	–31.03	–56.12	0	8	–	–	–	–	–	–
UR32**(34)	–31.26	–55.83	6	8	23.9	229	63	8.5	–41.3	202.9
UR33 (35)	–31.86	–56.22	4	8	–37.5	5.2	153	7.5	–78.2	148.2

The mean directions are in reasonably good agreement with the expected paleodirections for Early Cretaceous, as derived from reference poles given by *Besse and Courtillot (2002)* for stable South America. The mean paleomagnetic pole of this study is shown in Fig. 5 and listed in Table 2 together with previously published PMP poles and a paleomagnetic pole from Africa with similar age rotated to the South America reference frame. In general, the pole obtained in this study, agrees reasonably well with other pole positions, in particular with southern PMP poles. The clear outlier is the pole reported by *Alva-Valdivia et al. (2003)* which may be attributed to local tectonic rotations or insufficient sampling to overcome the paleosecular variation. It should be noted that our results contradict the finding of *Ernesto et al. (1990)* regarding a 10° clockwise rotation. As also discussed by *Ernesto et al. (1999)* the PMP poles are somewhat different, indicating unrecognized tectonic disturbances. As a whole, the PMP poles are significantly different from the pole position suggested by hotspot reconstruction (*Muller et al., 1993*) which may be due to true polar wander or hotspot motion.

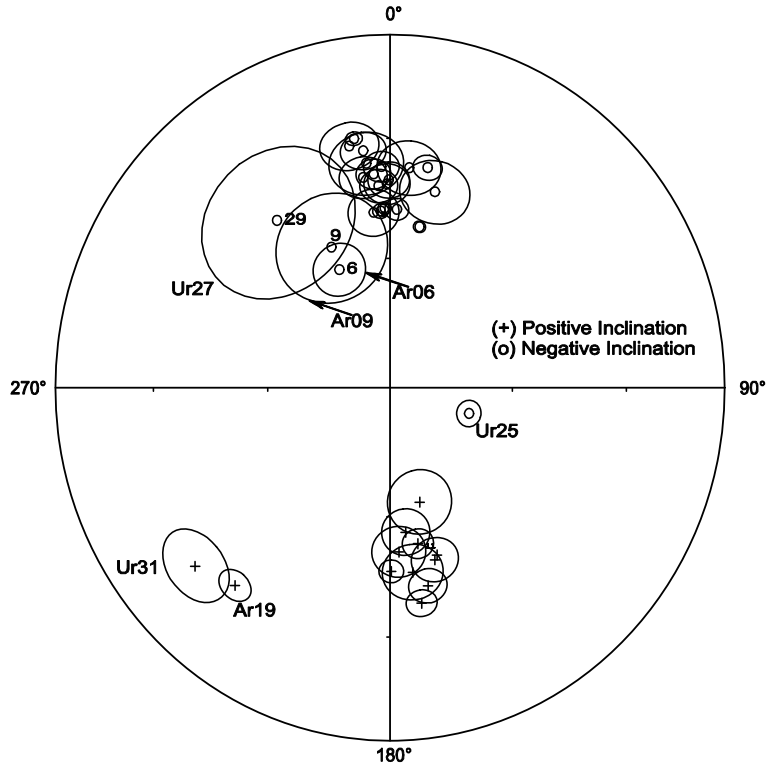


Fig. 4. Equal area projections of the flow mean characteristic paleodirections for all studied flows.

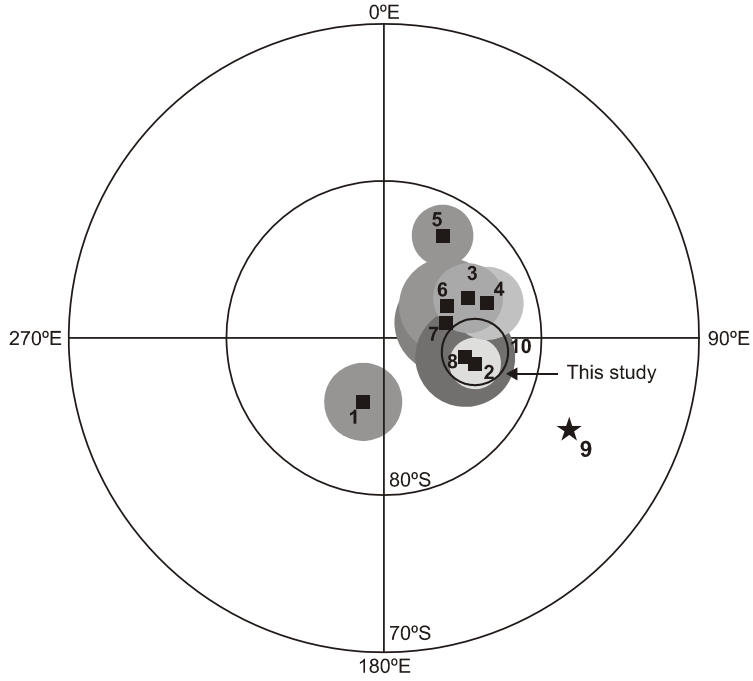


Fig. 5. The paleomagnetic pole for this study together with other paleomagnetic poles of the same age listed in Table 2. Numbering of the poles is the same as in Table 2. Fisher 95% confidence limits are also shown. Star is for the South American paleomagnetic pole derived from fixed hotspot rotations (*Muller et al., 1993*).

The formula $S_B^2 = S_T^2 - \left(S_W^2 / n\right)$ was used to estimate the paleosecular variation, where, S_T is the total angular dispersion $S_T = \left[1/(N-1)\right]\sqrt{\sum_{i=1}^{N}\delta_i^2}$ (*Cox, 1969*), N is the number of sites used in the calculation, δ_i the angular distance of the i-th virtual geomagnetic pole (VGP) from the axial dipole, S_W is the site dispersion (following *McEllhinny and McFadden, 1997*) and n is the average number of sample per site. As showed by *Biggin et al. (2008)* the commonly accepted calculation of the internal dispersion may be affected by some artifact. Thus, in this study, we use the *Biggin et al. (2008)* approach to calculate the internal dispersion adopting 45° cut-off angle (*Johnson et al., 2008*) to separate paleosecular variation and intermediate geomagnetic regimes. So, we obtained $S_B = 11.8$ with $S_U = 14.3$ and $S_L = 9.98$ (upper and lower limit, respectively), which agrees well with the selected data reported for the Cretaceous Normal Superchron (Fig. 6). A similar comparison with the Jurrasic data is quite delicate due to the limitation of paleolatitudinal span and relatively poor quality of available data. Our data reinforces the hypothesis outlined by *Biggin et al. (2008)* about the different style of secular

Table 2. Selected South American paleomagnetic poles for 120–140 Ma. *N* is a number of sites used for determination; Lat. and Long. are the latitude and longitude of the paleomagnetic pole; α_{95} is the confidence cone of the Fisher statistics. *ASD* is the estimated angular standard deviation. See text for more details.

Sampling Location	Age [Ma]	*N*	Lat. [°N]	Long. [°E]	α_{95} [°]	*ASD* [°]	Reference
Central PMP	~ 132	35	–85.7	197.9	2.6	8.5+1.7/–1.2	*Alva-Valdivia et al. (2003)*
Southern PMP	~ 133	197	–84.0	106.2	1.5	12.1+0.9/–0.8	*Raposo and Ernesto (1995)*
Central PMP	~ 132	103	–84.1	64.4	2.3	13.1+1.4/–1.2	*Raposo and Ernesto (1995)*
Northern PMP	~ 132	92	–83.0	71.4	2.4	13.0+1.5/–1.2	*Ernesto et al. (1999)*
Ponta Grossa	~ 130	115	–82.4	30.3	2.0	12.2+1.2/–1.0	*Raposo and Ernesto (1995)*
Alkaline Province Paraguay	127-130	75	–85.4	62.3	3.1	15.0+1.9/–1.5	*Ernesto et al. (1996)*
Cordoba Province Argentina	~ 125	55	–86.0	75.9	3.3	–	*Geuna and Vizan (1998)*
Kaoko basalts*	~ 132		–84.7	103.9	3.2	–	*Gidskehaug (1975)*
Fixed hotspot**	~ 130	-	–76.7	116.4	-	–	*Muller et al. (1993)*
Arapey	~ 132	29	–84.8	95.8	4.2	11.8+2.5/–1.8	This Study

* Rotated to South America from Africa (*Geuna et al., 2000*).

** Location of South American paleomagnetic pole predicted from fixed hotspot rotations (*Muller et al., 1993*).

variation during (and before) CNS and Plio-Quaternary (Fig. 6) supporting the link between PSV and reversal frequency.

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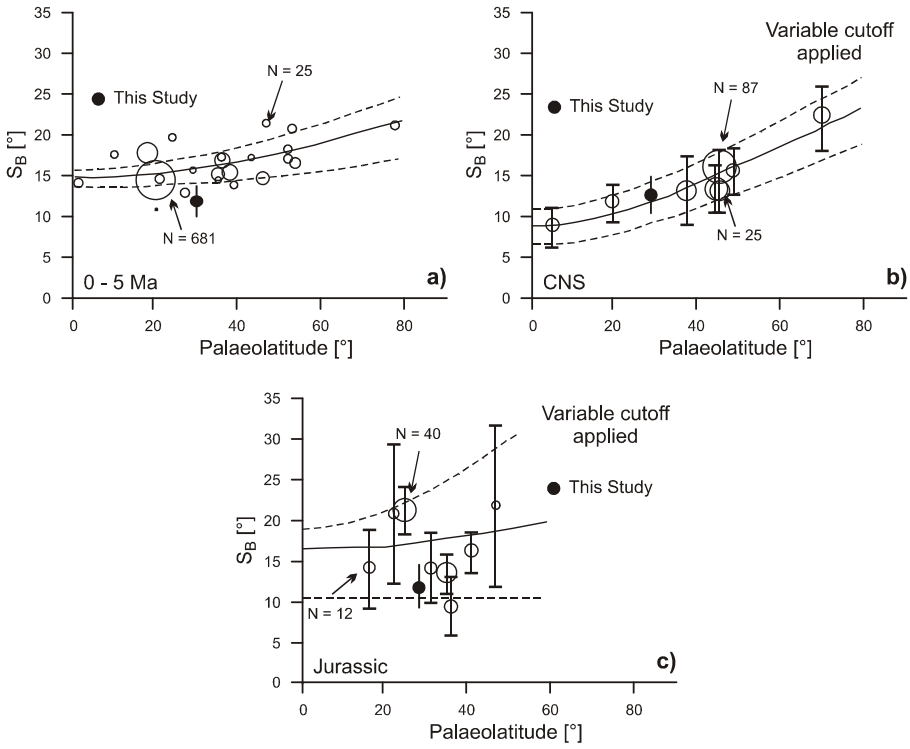


Fig. 6. Angular standard deviation of virtual geomagnetic poles (calculated using *Biggin's (2008)* approach) against palaeolatitude for: **a)** Pliocene, **b)** CNS, and **c)** Jurassic periods.

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